

Western Mathematics

TOPIC REVISION QUESTIONS

Mathematics Standard 2

MS-S1 Data Analysis

Section I – Multiple Choice Questions (1 Mark each)

Section II – Longer Questions (Marks indicated beside questions)

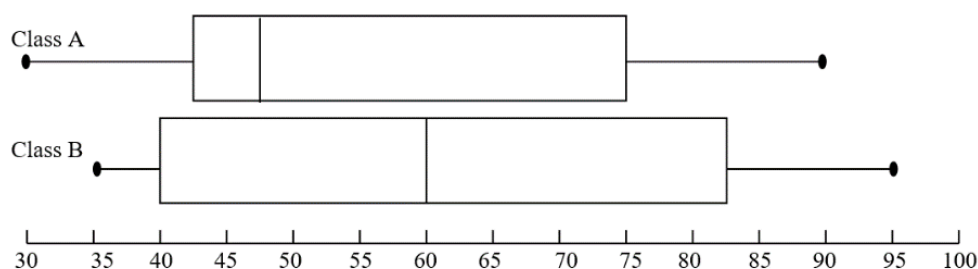
Section I**Multiple Choice Questions (1 Mark each)****15 marks**

1. The scores 16, 12, 11, 11, 6, 8, 10 have a range of:
 - (A) 11
 - (B) 10.6
 - (C) 10
 - (D) 7

2. The scores below are the marks that Julio scored on 10 skills tests.
12, 14, 20, 20, 25, 32, 32, 34, 38, 40
The interquartile range of the scores is:
 - (A) 6
 - (B) 19
 - (C) 14
 - (D) 28

3. In a survey, Reuben is asked to give some information about himself. The answers to which of following questions would provide data which is quantitative and discrete?
 - (A) Eye colour
 - (B) Gender (M or F)
 - (C) Height
 - (D) Number of siblings

4. Alex works in an electrical store and asks each customer who purchases an item their postcode. Which type of data is Alex collecting?
- (A) Categorical Ordinal
(B) Categorical Nominal
(C) Quantitative Continuous
(D) Quantitative Discrete
5. Leica collects information from the members of her netball team, which she needs to order uniforms. Which is an example of ordinal categorical data?
- (A) Their waist measurements in mm.
(B) Their height in cm.
(C) The colour of shirts they want.
(D) Their shirt sizes (S, M, L, XL....)
6. The results gained by the students in two classes A and B on a test are shown below:



Which statement is true?

- (A) Both classes had the same interquartile range.
(B) Both classes had the same maximum score.
(C) Both classes had the same median score.
(D) Both classes had the same range.

7. The points scored by a rugby team over 14 games is shown in the stem and leaf plot below.

Stem	Leaf
0	6 9 9
1	0 2 5 6 8 7
2	2 5 8 9
3	3

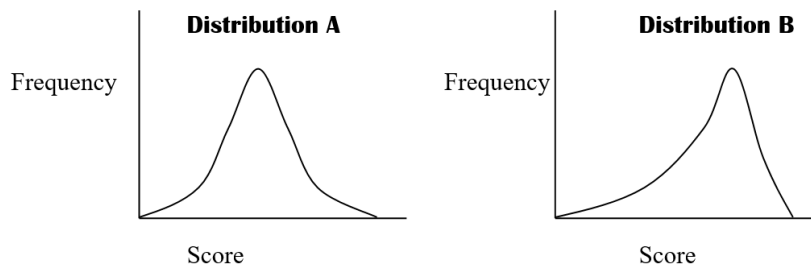
What is the interquartile range of the scores?

- (A) 17
(B) 15
(C) 25
(D) 5
8. Which is not true of the following set of scores?

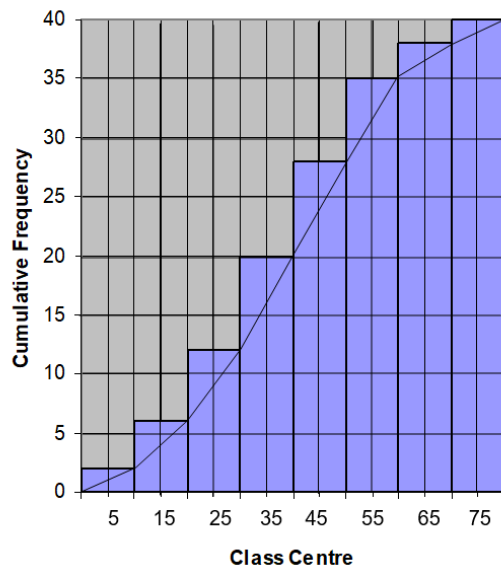
Score	Frequency
12	3
13	9
14	5
15	5
16	3
Total	25

- (A) The mean is 13.84.
(B) The median is 13.
(C) The mode is 13.
(D) The range is 4.

9. Which best describes the two distributions shown below.



- (A) Both distributions are symmetrical.
- (B) Distribution A is symmetrical and B is positively skewed.
- (C) Distribution A is symmetrical and B is negatively skewed
- (D) Both distributions are negatively skewed.
10. A cumulative frequency histogram and polygon is shown for a test out of 80 marks attempted by 40 students.



The interquartile range is closest to:

- (A) 30
- (B) 40
- (C) 50
- (D) 60

11. Sven throws three darts at a dartboard fifteen times and records his scores.
After a visit to the spa Sven repeats the experiment.
The results are shown in the stem and leaf plot below.

Before spa			After spa		
9	1	2			
9	8	2	3	5	6
9	7	1	4	0	5
9	6	0	5	1	3
6	6	1	6	1	2
	2	7		1	6
		8		2	8

Which statement is true?

- (A) Both the mean and the range have decreased.
 (B) Both the mean and the range have increased.
 (C) The mean has decreased, but the range has increased.
 (D) The mean has increased, but the range has decreased.
12. Which is correct for the set of data below, correct to the nearest whole number?

Score (x)	Frequency (f)	fx
12	3	36
13	5	65
14	7	98
15	8	120
16	2	31

$$\Sigma f = 25$$

$$\Sigma fx = 350$$

- (A) The mean is 14 and the mode is 15.
 (B) The mean is 15 and the mode is 14.
 (C) The mean is 14 and the mode is 14.
 (D) The mean is 15 and the mode is 15.

13. The marks from 10 people on a test are given below.

13, 16, 30, 21, 24, 18, 17, 23, 30 and 28

Which statement is correct?

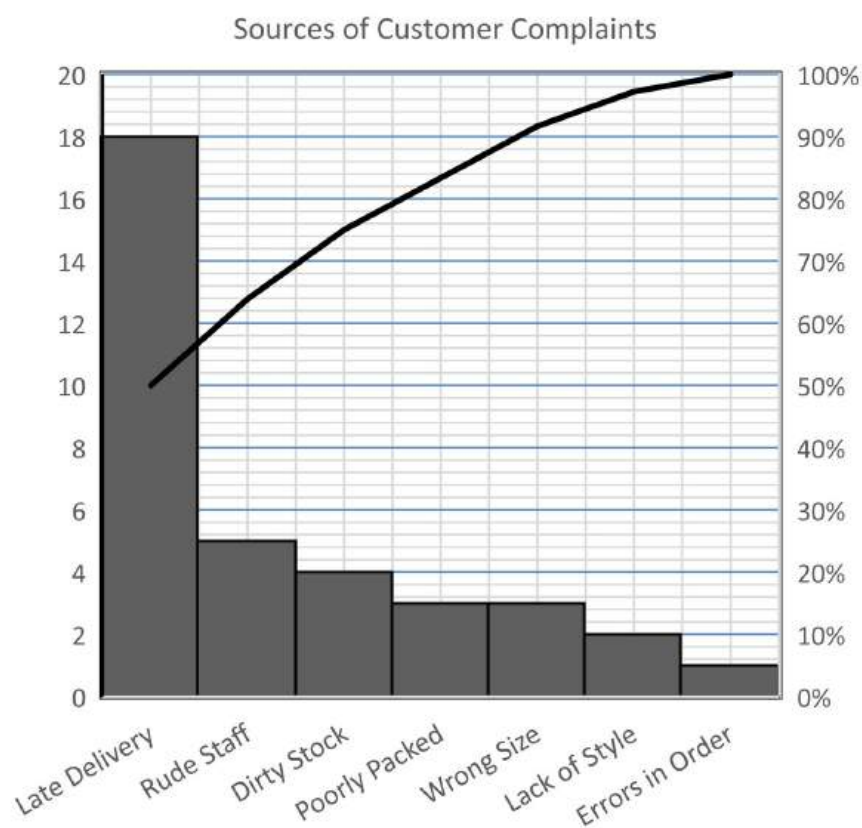
- (A) The mean and median are both 22 and the mode is 30.
 - (B) The mean is 22, the median is 23 and the mode is 30.
 - (C) The mean is 22, the median is 30 and the mode is 21.
 - (D) The mean and the mode are both 22 and the median is 30.
14. There are 48 people at a party. The ages of a sample of 8 people are listed below.

15, 15, 16, 13, 14, 12, 17, 16

The standard deviation, correct to three significant figures, is :

- (A) 1.56
- (B) 1.67
- (C) 2.44
- (D) 14.8

15. The Pareto Chart below was constructed to analyse the complaints received by a phone and online clothing retailer.



What percentage of complaints related to Rude Staff?

- (A) 11%
- (B) 14%
- (C) 50%
- (D) 64%

Western Mathematics

Mathematics Standard 2

MS-S1 Data Analysis

REVISION QUESTIONS - Section II

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 16 (6 marks)

Marks

The table represents the total number of goals scored in a series of soccer matches.

Number of Goals	<i>frequency</i>
0	14
1	10
2	12
3	5
4	4
5	2
6	1

(a) What was the total number of matches played?

1

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(b) Complete a 5-figure summary for this data.

2

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(c) Draw an accurate box-plot of this data.

2



(d) Describe the shape of the distribution in terms of skew.

1

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Question 17 (7 marks)

The results of a class test are shown on the stem and leaf plot below.

Stem	Leaf
1	7 8 8
2	2 3 5 7 9 9
3	2 2 2 4 6 8 9
4	0 2 5

- (a) What is the median mark? 1

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- (b) What is the interquartile range? 2

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- (c) Determine if there are any outliers in the data (give reasoning and calculations to justify your answer). 2

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- (b) What is the mean? 1

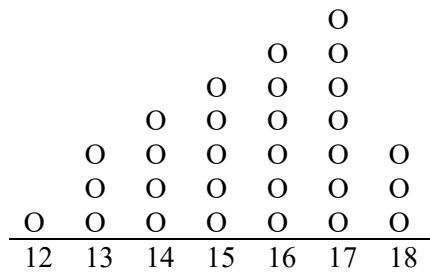
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- (c) What is the population standard deviation? 1

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Question 18 (5 marks)

A dot plot is shown below of the ages of 29 people at a party.



- (a) Find the range of the ages.

1

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- (b) Find the interquartile range of the ages.

2

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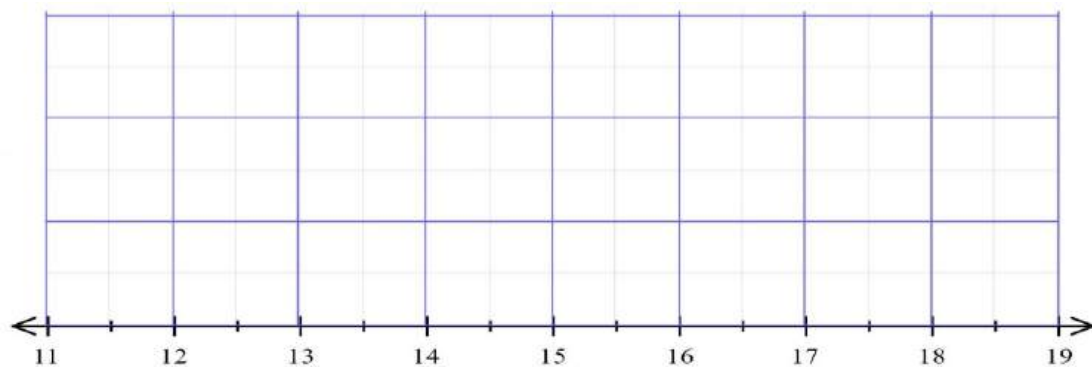
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- (c) Draw a box and whisker plot to represent the data.

2

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Question 19 (4 marks)

A School of 1 200 students has the following year groups.

School Year	7	8	9	10	11	12
Number of students	250	260	200	190	160	140

Each year is approximately evenly split in the gender of students. A stratified random sample of 60 students is to be chosen for a survey.

(a) How many Year 9 students should be included in the sample?

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(b) How many Year 11 girls should be included in the sample?

2

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Question 20 (5 marks)

Basketball NSW conducts a survey of its players across the state, using a sample of 250 players.

- (a) If the Western Region has 8% of the basketball players in NSW, how many players from the Western Region should be included in the sample? **1**

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- (b) The stem and leaf plot below shows the results for the question on the heights (in cm) of the 22 players from the Southern Region. **3**

16	6 7 9
17	2 4 8 8 9
18	3 4 5 5 8 9 9
19	0 3 6 8 8
20	0 2

Draw a box-plot from this data.

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- (c) Find the standard deviation of this sample to 2 decimal places. **1**

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Question 21 (4 marks)

A group of 20 people are asked to rate a TV program on a scale from 1 to 10. The results are shown below.

1	7	9	4	8
6	5	5	4	7
8	10	6	7	4
6	7	8	7	8

(a) Draw a dot plot of the data above.

2

(b) Give the mode and the median of the data.

2

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Question 22 (5 marks)

The frequency distribution table below is of the ages of students in a club at school.

Age	Frequency	Cumulative Frequency
12	5	
13	14	
14	16	
15	17	
16	12	
17	14	
18	10	

- (a) Calculate the mean and standard deviation of the ages.

2

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- (b) Complete the Cumulative frequency column, on the table above.

1

- (c) Find the median and interquartile range.

2

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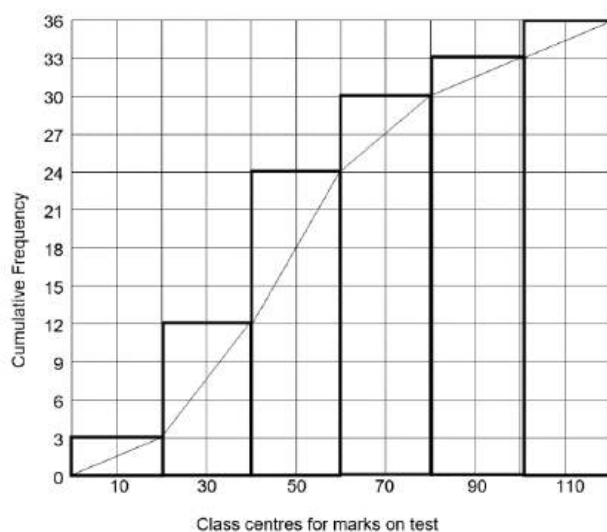
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Question 23 (3 marks)

A Cumulative Frequency Histogram and Polygon for the marks on a test is shown below.



By drawing lines on the graph, find the five number summary for the data.

3

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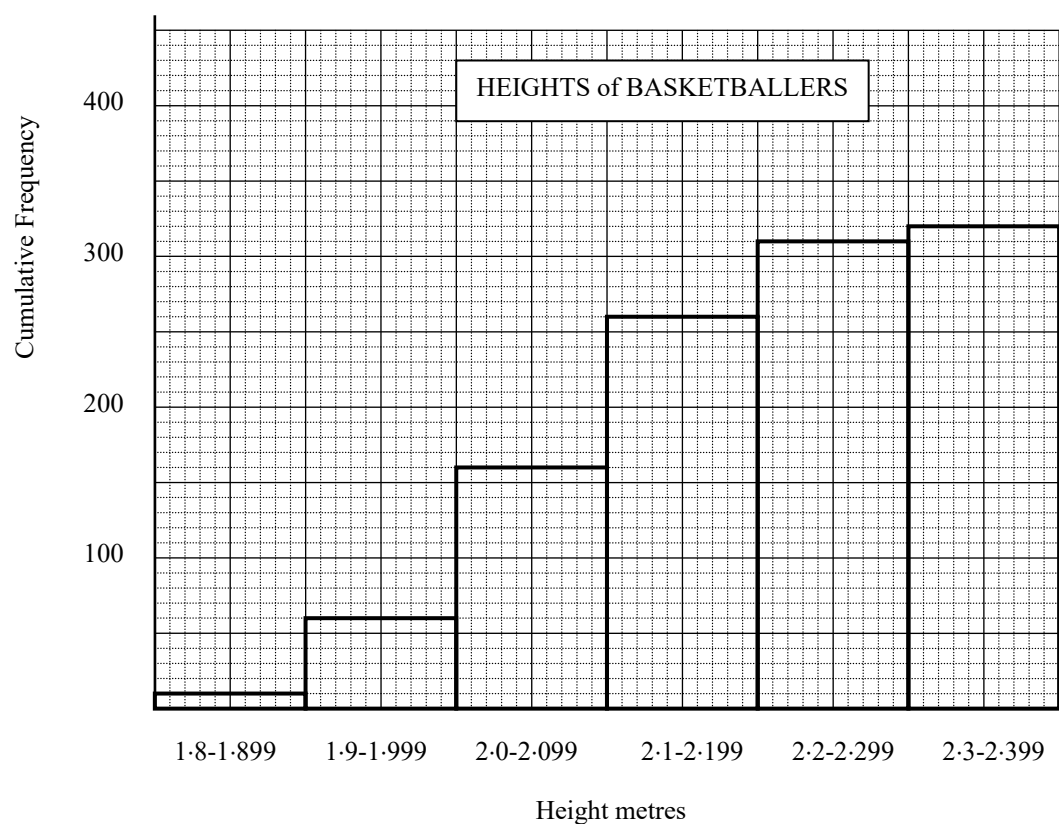
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Question 24 (5 marks)

An administrator has been given the following information about players in a league they manage.



- (a) How many players have a height between 2.0 and 2.199m?

1

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- (b) Complete a Cumulative Frequency Polygon on the diagram above.

2

- (c) Use the polygon to obtain the values for a five-figure-summary for this data.

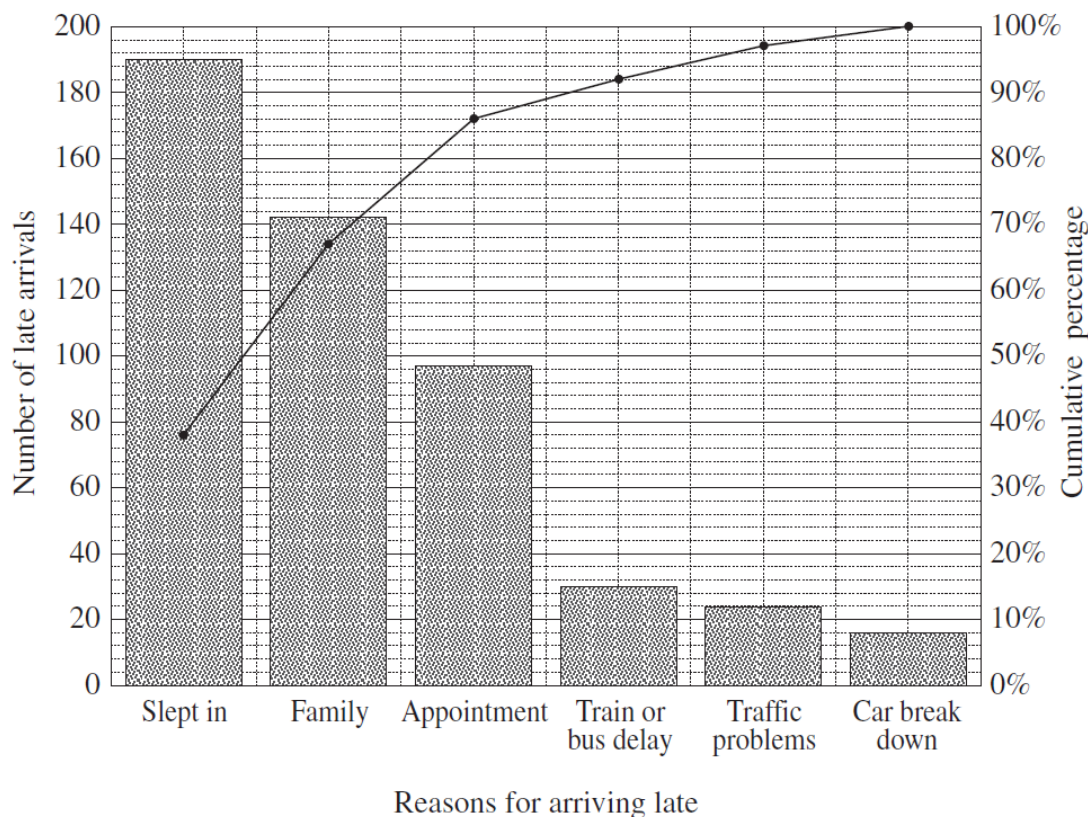
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Question 25 (3 marks)

A school collected data related to the reasons given by students for arriving late. The Pareto chart shows the data collected.



- (a) What percentage of the students gave the reason “Family”?

1

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- (b) Which single reason accounted for approximately 20% of lateness?

1

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- (c) If they try to address the issues which account for at least 80% of the lateness, which issues should they target?

1

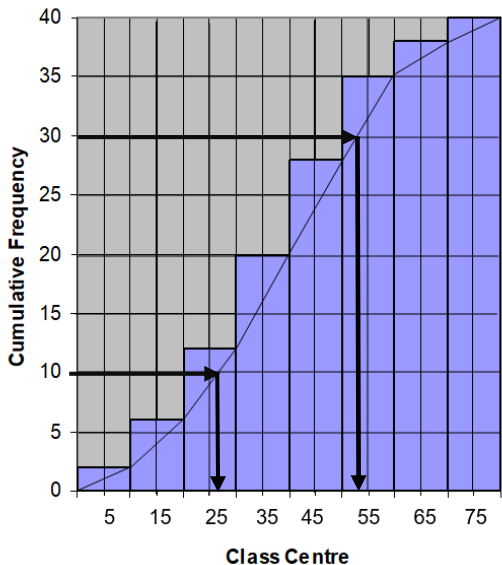
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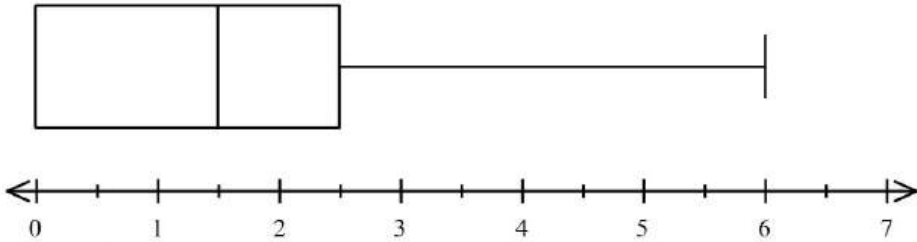
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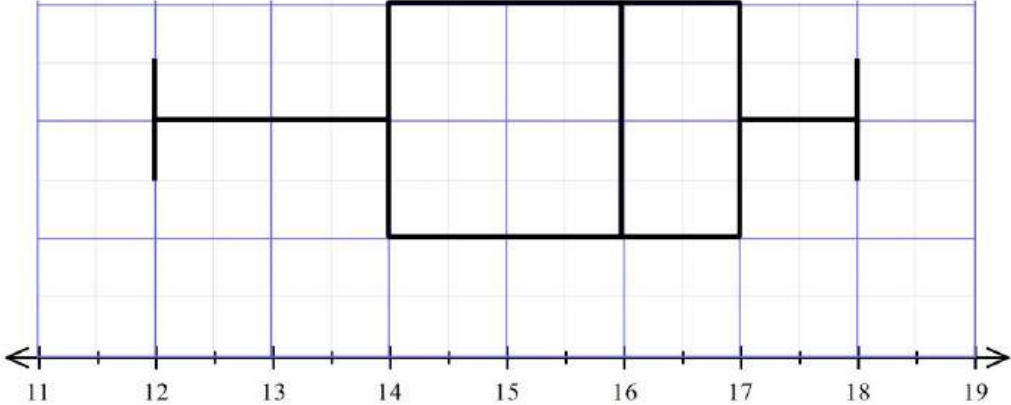
Section I - Worked Solutions

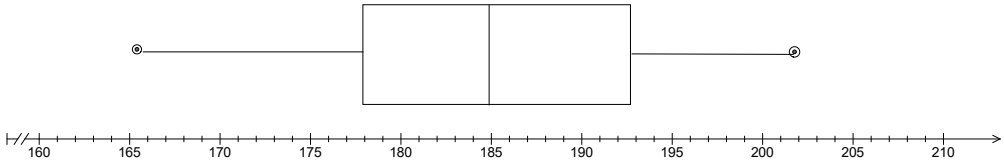
No	Working	Answer															
1	Range = 16 – 6 = 10	C															
2	12, 14, 20, 20, 25, 32, 32, 34, 38, 40 $Q_2 = \frac{25 + 32}{2} = 28.5$ $Q_1 = 20 \qquad Q_3 = 34$ $IQR = 34 - 20 = 14$	C															
3	Quantitive data involves numbers, and discrete data are numbers from counting or otherwise separate numbers, so the “Number of siblings” meets this criteria.	D															
4	A postcode (although it is a number) is neither counted or measured, so is a categorical data item and (although it is a number) does not indicate any implied order, so it is Categorical Nominal data.	B															
5	D The shirt sizes are names , so are Categorical, but they do imply an order so are Ordinal categorical. A and B are measurements, so are Quantitative Continuous, and C is Categorical Nominal	A															
6	<table><tr><td></td><td>Class A</td><td>Class B</td></tr><tr><td>IQR</td><td>75 – 42 = 33</td><td>82 – 40 =42</td></tr><tr><td>Max Score</td><td>90</td><td>95</td></tr><tr><td>Median</td><td>47</td><td>60</td></tr><tr><td>Range</td><td>90 – 30 = 60</td><td>95 – 35 = 60 (SAME)</td></tr></table>		Class A	Class B	IQR	75 – 42 = 33	82 – 40 =42	Max Score	90	95	Median	47	60	Range	90 – 30 = 60	95 – 35 = 60 (SAME)	D
	Class A	Class B															
IQR	75 – 42 = 33	82 – 40 =42															
Max Score	90	95															
Median	47	60															
Range	90 – 30 = 60	95 – 35 = 60 (SAME)															
7	<table><tr><td>Stem</td><td>Leaf</td></tr><tr><td>0</td><td>6 9 9</td></tr><tr><td>1</td><td>0 2 5 6 8 7</td></tr><tr><td>2</td><td>2 5 8 9</td></tr><tr><td>3</td><td>3</td></tr></table> 14 scores divide evenly into 2 sets of 7, so find 4th in each set. IQR = 25 – 10 = 15	Stem	Leaf	0	6 9 9	1	0 2 5 6 8 7	2	2 5 8 9	3	3	B					
Stem	Leaf																
0	6 9 9																
1	0 2 5 6 8 7																
2	2 5 8 9																
3	3																

8	Mean = 13.84, from calculator Median = 14 (13 th score out of 25) So median = 13 is incorrect Mode = 13 (most common score, highest frequency) Range = 16 – 12 = 4			B
9	A is symmetrical and B is skewed with the tail pointing downward, so negatively skewed			C
10	 IQR = 54 – 26 = 28 So closest to 30			A
11	The mean does not need to be calculated, the increase in scores can be seen from the plot so mean has increased. The calculated ranges show that it has decreased	Before Spa Mean = 48.4 Range = 72 – 21 = 51	After Spa Mean = 55.4 Range = 82 – 35 = 47	D
12	Mean = $\frac{\Sigma fx}{\Sigma f} = \frac{350}{25} = 14$ Mode = 15 (highest frequency)			A
13	In order – 13, 16, 17, 18, 21, 23, 24, 28, 30, 30 Mean = $\frac{\Sigma fx}{\Sigma f} = \frac{220}{10} = 22$ Mode = 30 (occurs twice) Median = $\frac{21 + 23}{2} = 22$ The mean and median are both 22 and mode is 30			A

14	From Calculator Mean = $\bar{x} = 14.75$ Sample stand deviation = $s = 1.669 = 1.67$ (3 sig fig)	B																																
15	Sources of Customer Complaints <table><caption>Data for Sources of Customer Complaints</caption><thead><tr><th>Source</th><th>Count</th><th>Cumulative Count</th><th>Cumulative Percentage</th></tr></thead><tbody><tr><td>Late Delivery</td><td>18</td><td>18</td><td>50.0%</td></tr><tr><td>Rude Staff</td><td>5</td><td>23</td><td>64.0%</td></tr><tr><td>Dirty Stock</td><td>4</td><td>27</td><td>75.0%</td></tr><tr><td>Poorly Packed</td><td>3</td><td>30</td><td>83.3%</td></tr><tr><td>Wrong Size</td><td>3</td><td>33</td><td>91.7%</td></tr><tr><td>Lack of Style</td><td>2</td><td>35</td><td>97.2%</td></tr><tr><td>Errors in Order</td><td>1</td><td>36</td><td>100.0%</td></tr></tbody></table> OR Total = $18 + 5 + 4 + 3 + 3 + 2 + 1$ = 36 % Rude Staff = $\frac{5}{36} \times 100 = 13.88888889 = 14\%$	Source	Count	Cumulative Count	Cumulative Percentage	Late Delivery	18	18	50.0%	Rude Staff	5	23	64.0%	Dirty Stock	4	27	75.0%	Poorly Packed	3	30	83.3%	Wrong Size	3	33	91.7%	Lack of Style	2	35	97.2%	Errors in Order	1	36	100.0%	B
Source	Count	Cumulative Count	Cumulative Percentage																															
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Wrong Size	3	33	91.7%																															
Lack of Style	2	35	97.2%																															
Errors in Order	1	36	100.0%																															

MS-S1 Data Analysis Section II Solution		Marks
Question 16		
(a)	$14 + 10 + 12 + 5 + 4 + 2 + 1 = 48$	1
(b)	$Q_0 = 0$ $Q_1 = 0$ Q_2 is average of 24 and 25 th scores = 1.5 Q_3 is average of 36 th and 37 th scores = 2.5 $Q_4 = 6$ Summary: 0, 0, 1.5, 2.5, 6	2
(c)		2
(d)	It has a positive skew.	1
Question 17		
(a)	From 19 scores the median Q_2 is the 10 th score = 32	1
(b)	Q_1 (5 th score) = 23 Q_3 (15 th score) = 38 $IQR = 38 - 23 = 15$	2
(c)	Outlier can be defined as outside $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$ $Q_1 - 1.5 \times IQR = 23 - 1.5 \times 16 = 0.5$ $Q_3 + 1.5 \times IQR = 38 + 1.5 \times 16 = 60.5$ There are no scores outside these limits so there are no outliers.	2
(d)	Mean = 30.4 (1 dec place)	1
(e)	SD = 8.2 (1 dec place)	1
Question 18		
(a)	Range = $18 - 12 = 6$	1

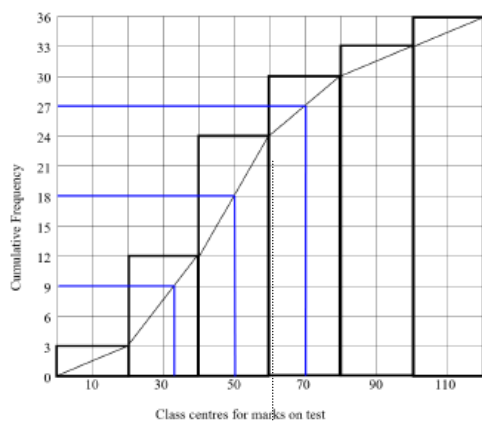
(b)	Median (15th score) = 16 Q_1 (7th and 8th scores) = 14 Q_3 (22nd and 23rd scores) = 17 Interquartile Range = $17 - 14 = 3$	2
(c)		2
Question 19		
(a)	i) Year nine are $\frac{200}{1200} = \frac{1}{6}$ of the school Number in sample = $\frac{1}{6} \times 60 = 10$ students	2
(b)	ii) Year 11 girls are $\frac{80}{1200} = \frac{1}{15}$ of the school Number in sample = $\frac{1}{15} \times 60 = 4$ girls	2
Question 20		
(a)	Number of players from Western = 8% of 250 = 20 players	1

(b)	Q_0(1st score) = 166 Q_1 (6th score) = 178 Q_2(11th and 12th) = 185 Q_3(17th score) = 193 Q_4(22nd score) = 202 	3
(c)	Sample SD = 10.85 (From Stats function on a calculator)	1

Question 21											
(a)	0 00 0000 00000 00000000 12345678910										2
(b)	Mode = 7 and median = 7										2
Question 22											
(a)	Mean = 15.125 (from calculator) Population Standard Deviation = 1.76										2
(b)		Age	Frequency		Cumulative Frequency						1
		12	5		5						
		13	14		19						
		14	16		35						
		15	17		52						
		16	12		64						
		17	14		78						
		18	10		88						
(c)	Median Q_2 (44 th and 45 th scores) =15 Q_1 (22 nd and 23 rd scores) = 14 Q_3 (66 th and 67 th scores) = 17 Interquartile range = 17 - 14 =3										2

Question 23

Five number summary 0, 34, 50, 70, 120



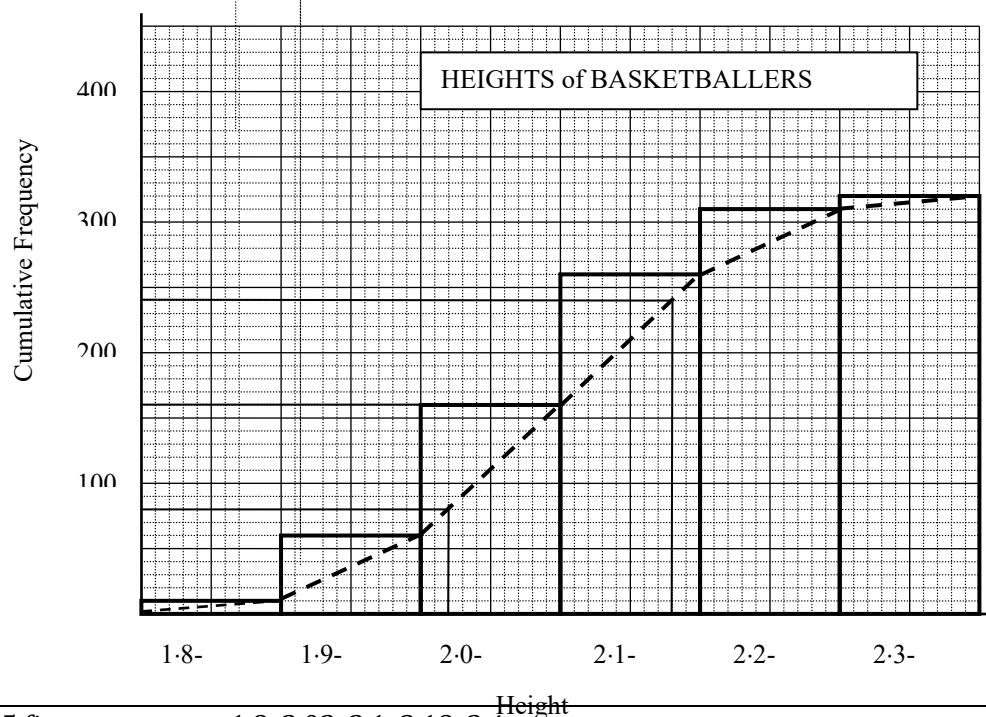
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Question 24

(a) 200

1

(b)

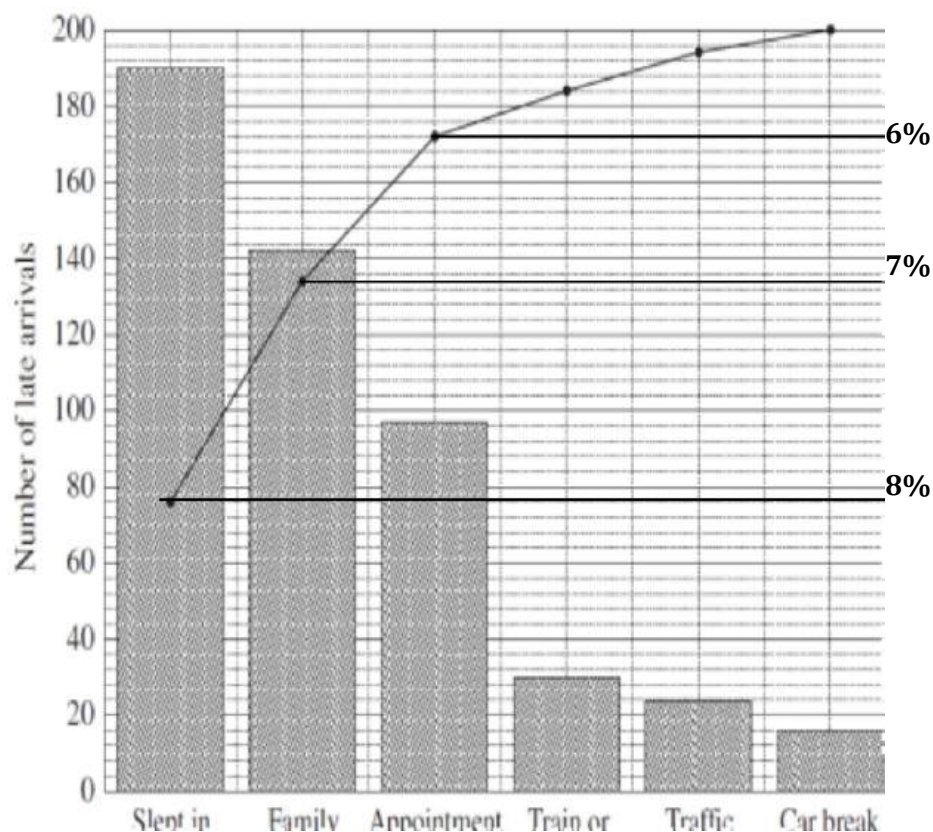


2

(c) The 5 figure summary 1.8, 2.02, 2.1, 2.18, 2.4

2

Question 25



(a)	Percentage "Family" = $67\% - 38\% = 29\%$	1
(b)	Percentage "Appointment" = $86\% - 67\% = 19\%$ Appointment is closest to 20%	1
(c)	Addressing the first two would account for 67% so to be over 80% we need to address first three, "Slept in", "Family" and "Appointment"	1